Project Title: Deployment of Ubuntu Server and Implementation of LVM on Ubuntu Server for Flexible Disk Management

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Deployment of Ubuntu Server and Implementation of LVM on Ubuntu Server for Flexible Disk Management

Introduction

In the modern world of IT infrastructure, server deployment and efficient storage management are key components for ensuring optimal performance, scalability, and flexibility. Ubuntu Server, a robust and widely-used Linux distribution, serves as a reliable platform for hosting services, applications, and databases. To complement this, Logical Volume Management (LVM) offers dynamic disk management, allowing system administrators to resize, manage, and allocate storage volumes with ease. This documentation outlines the deployment of Ubuntu Server in a virtual environment and the implementation of LVM to manage disk storage effectively.

What is Ubuntu Server?

Ubuntu Server is a version of the Ubuntu operating system specifically designed for server environments. Unlike the desktop version, it does not come with a graphical user interface by default, making it more lightweight and resource-efficient. Ubuntu Server is commonly used for hosting web applications, databases, file storage, and virtual machines. It is known for its security, stability, long-term support (LTS) releases, and ease of use, making it a popular choice among both beginners and experienced system administrators.

What is LVM (Logical Volume Management)?

LVM stands for Logical Volume Management. It is a flexible disk management system used in Linux environments. Unlike traditional disk partitioning, LVM allows you to create logical volumes that can span across multiple physical disks. These logical volumes can be resized or moved without affecting data, providing greater flexibility in managing storage.

Key Concepts:

- **Physical Volume (PV):** The actual hard disk or partition prepared for LVM use.
- **Volume Group (VG):** A pool of storage created by combining one or more physical volumes.
- **Logical Volume (LV):** A virtual partition created from a volume group that can be formatted and mounted.

Why Use LVM?

- Resize volumes easily without downtime.
- Combine multiple disks into a single logical volume.
- Simplify storage expansion and backup.

Prerequisites

Before deploying Ubuntu Server and implementing LVM, ensure the following requirements are met:

- **System Requirements:**
  - At least 2 GB of RAM.
  - 25 GB or more of available disk space.
  - Ubuntu Server ISO file (e.g., ubuntu-22.04.5-live-server-amd64.iso).
- **Software Requirements:**
  - VMware Workstation or VMware Player installed.
  - LVM2 package (usually included in modern Ubuntu installations).
- **Technical Knowledge:**
  - Basic understanding of Linux commands.
  - Familiarity with virtual machine configuration and Linux file systems.

Objective:

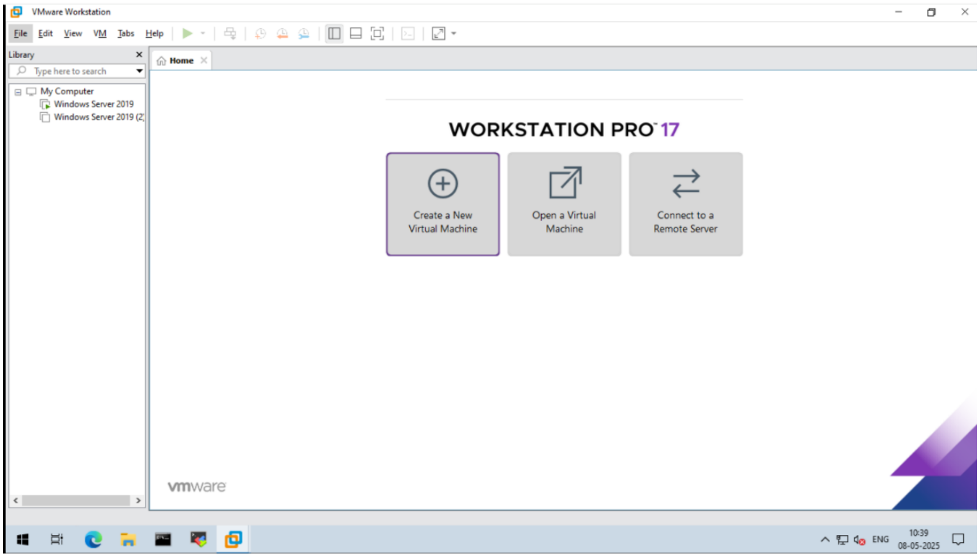
The primary objective of this documentation is to:

- Deploy Ubuntu Server in a virtual machine using VMware for a controlled and secure environment.
- Implement Logical Volume Management (LVM) by:
  - Adding an extra virtual hard disk.
  - Creating a physical volume.
  - Creating a volume group.
  - Dividing the volume group into multiple logical volumes.
  - Formatting and mounting each logical volume for use.
- Understand the benefits of LVM in real-world system administration tasks such as storage resizing and flexible data management.

Ubuntu Server Installation – Step-by-Step Process:

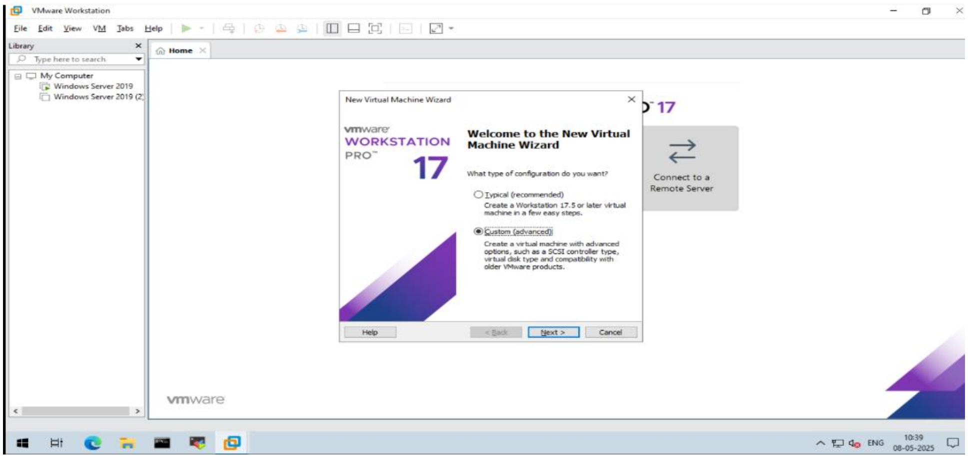
Step 1: Start VMware

- Open **VMware Workstation** or **VMware Player**.
- Click **Create a New Virtual Machine**.



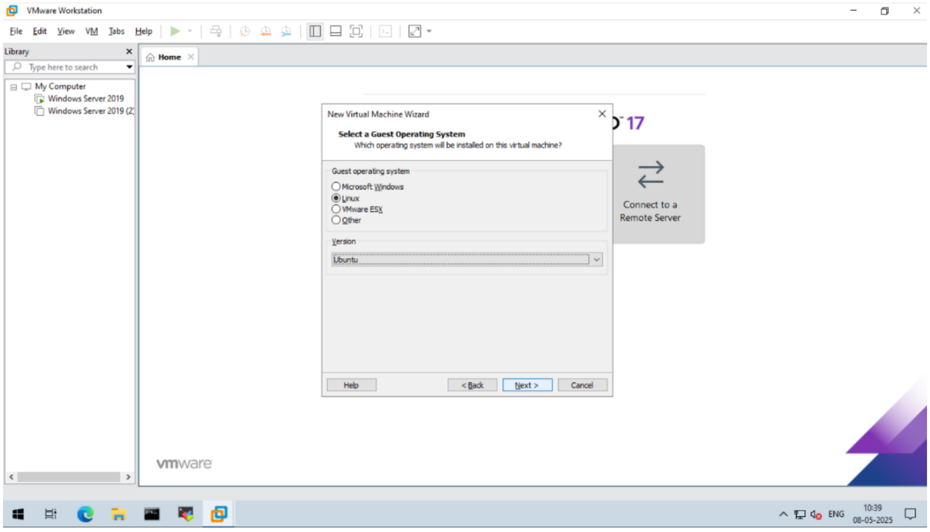
Step 2: Choose Configuration

- Select **Custom (advanced)**.
- Click **Next**.



Step 3: Select Guest Operating System

- Select **Linux** from the list.
- Click **Next**.

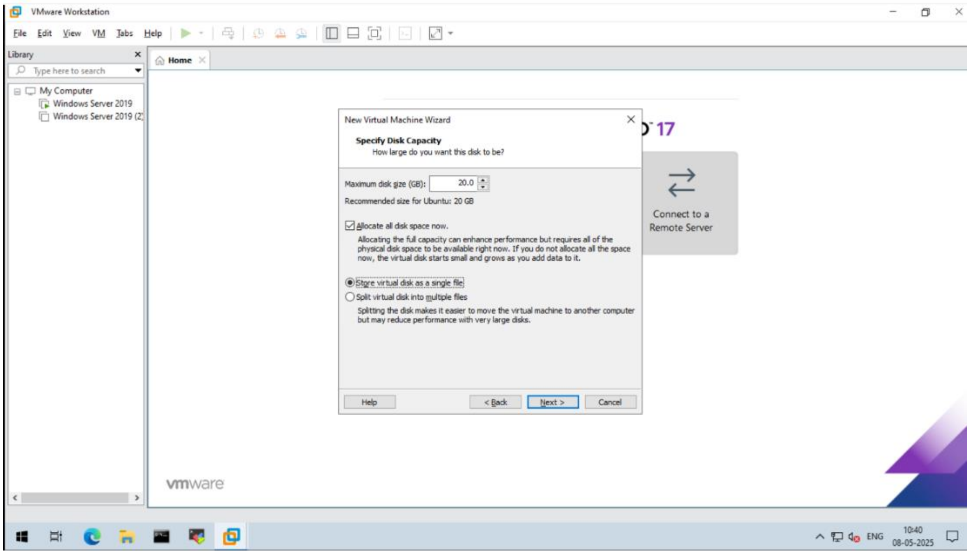


Step 4: Proceed Through Defaults

- Keep clicking **Next** until you reach the **Specify Disk Capacity** tab.

Step 5: Set Disk Configuration

- Select **Allocate all disk space now**.
- Choose **Store virtual disk as a single file**.
- Click **Next**.

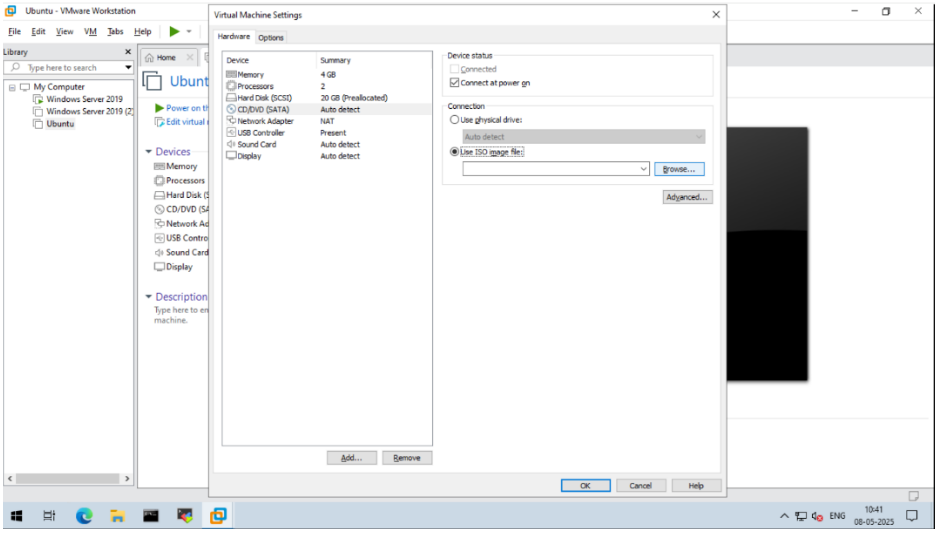


Step 6: Finish Initial Setup

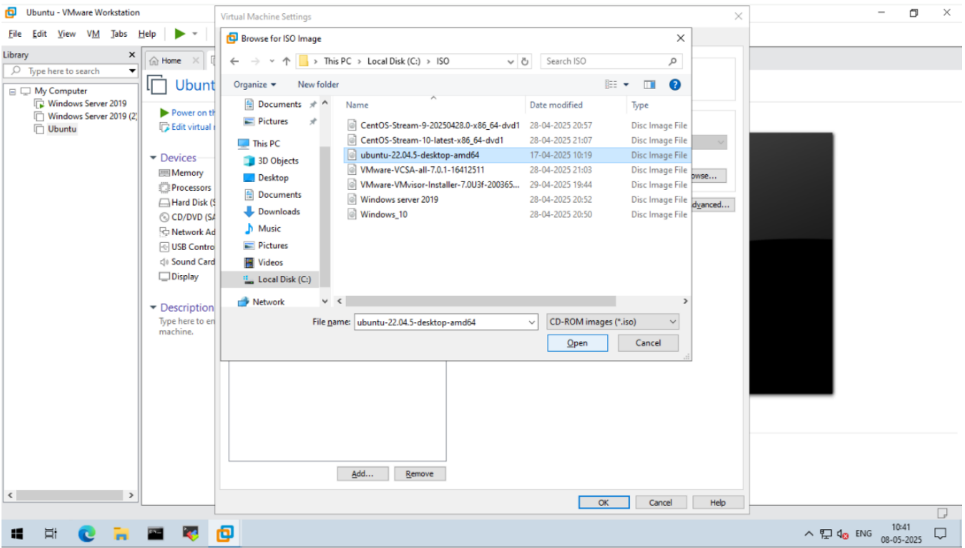
- Keep clicking **Next** until you see the **Finish** tab.
- Click **Finish**.

Step 7: Attach Ubuntu ISO File

- Click **Edit Virtual Machine Settings**.
- Select **CD/DVD (SATA)**.

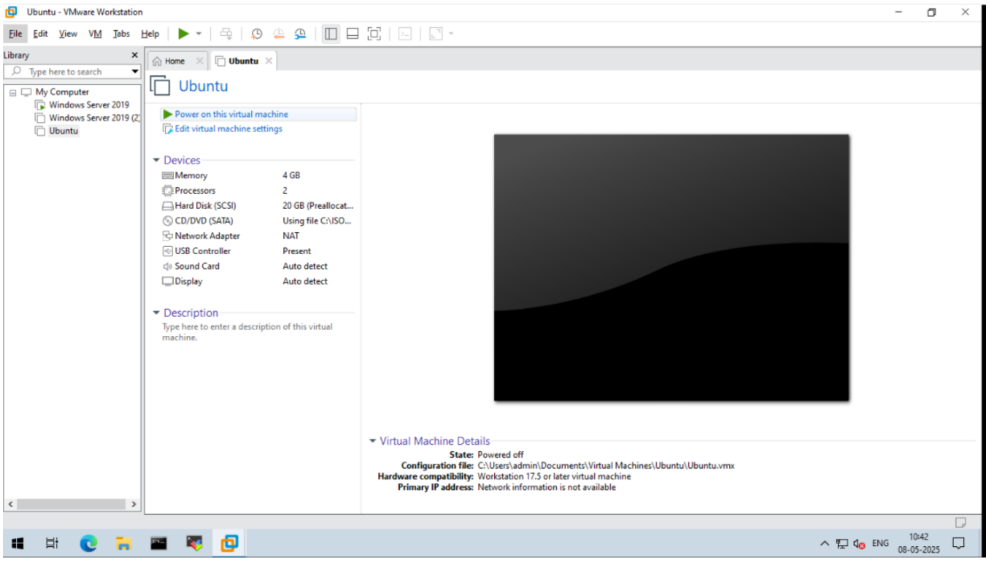


- Choose **Use ISO Image File**.
- Click **Browse**, go to C:\ISO\, select **Ubuntu-22.04.5-Desktop-AMD64.iso**, and click **Open**.
- Click **OK**.

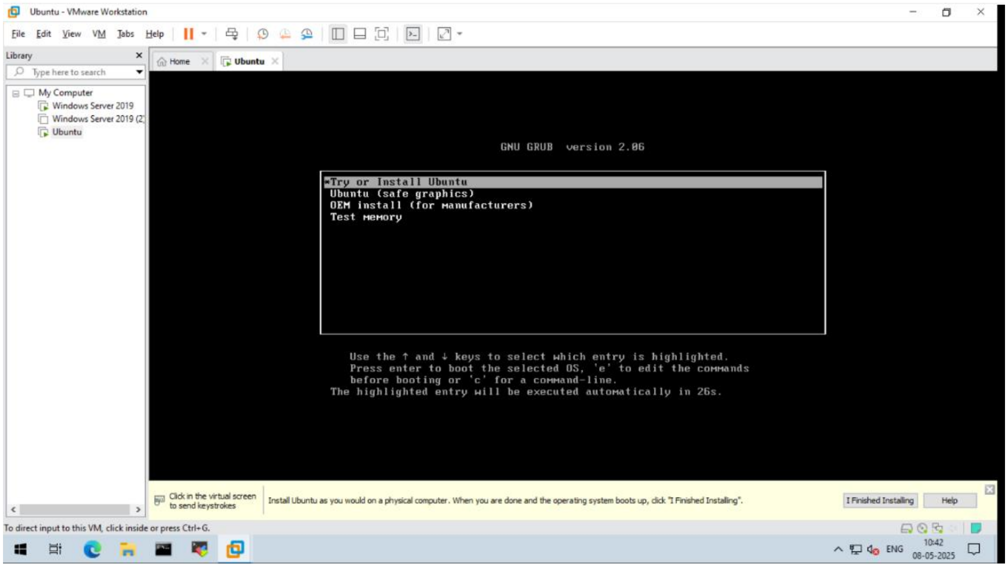


Step 8: Power On the Virtual Machine

- Click **Power on this Virtual Machine**.

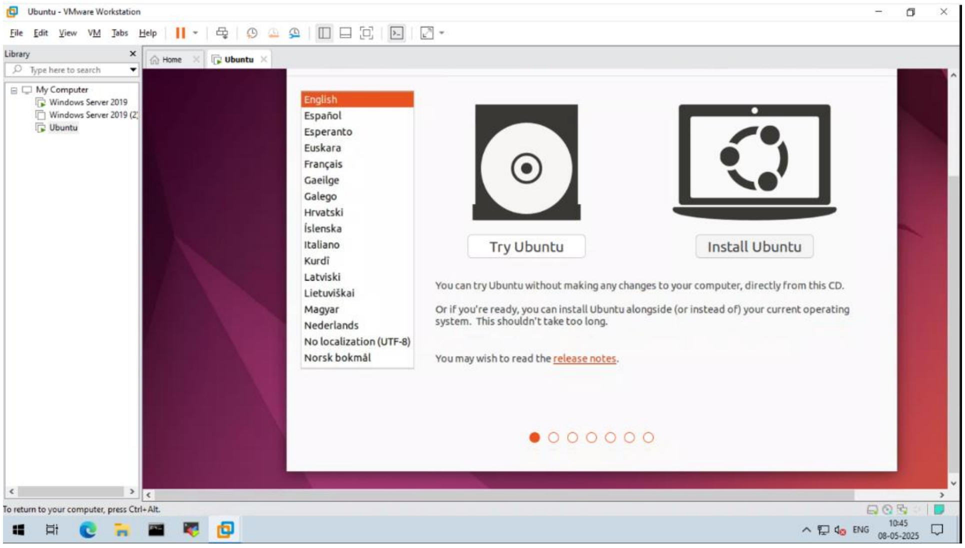


- On the **GNU GRUB v2.06** screen, select **Try or Install Ubuntu**.



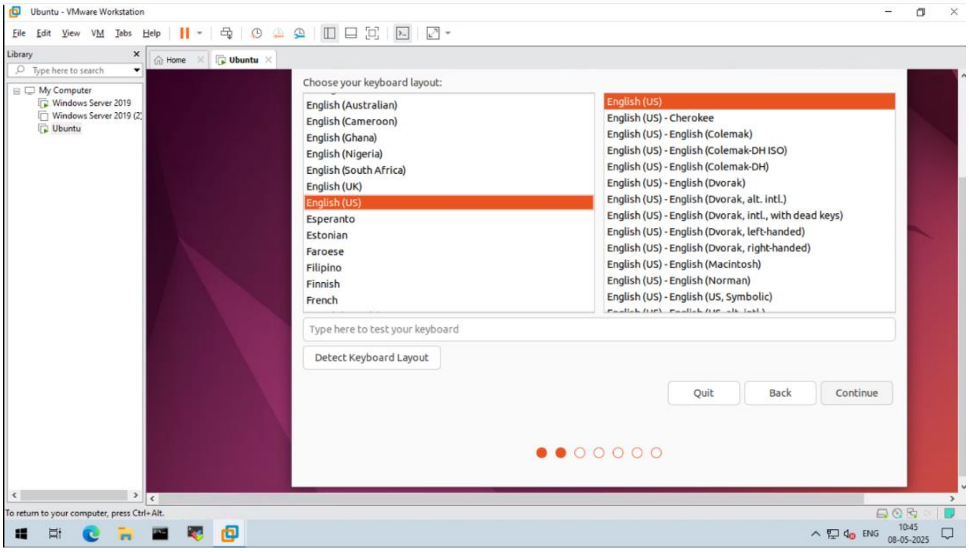
Step 9: Begin Ubuntu Installation

- Click **Install Ubuntu**.



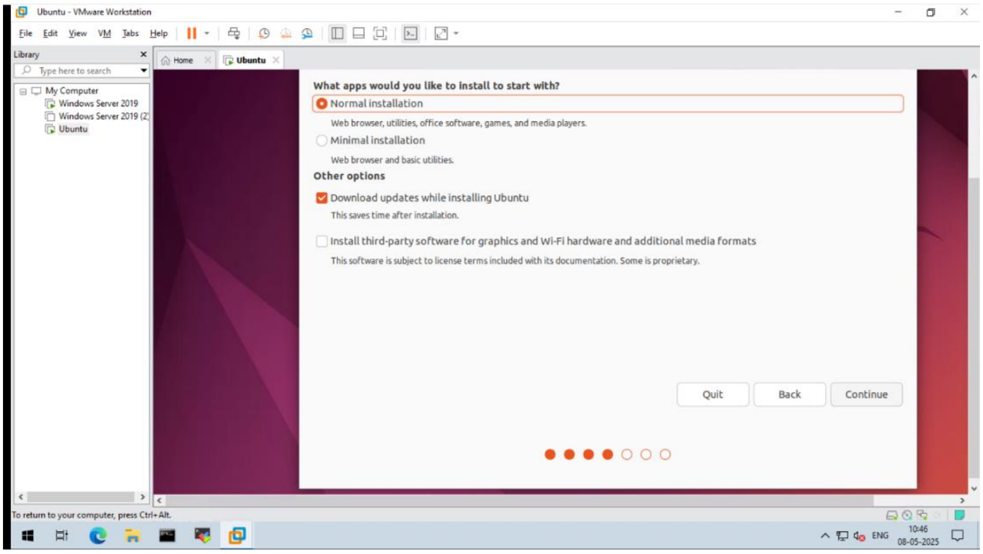
Step 10: Select Language

- Choose your preferred language.
- Click **Continue**.



Step 11: Installation Type

- Select **Normal Installation**.
- Check the box **Download updates while installing Ubuntu**.
- Click **Continue**.

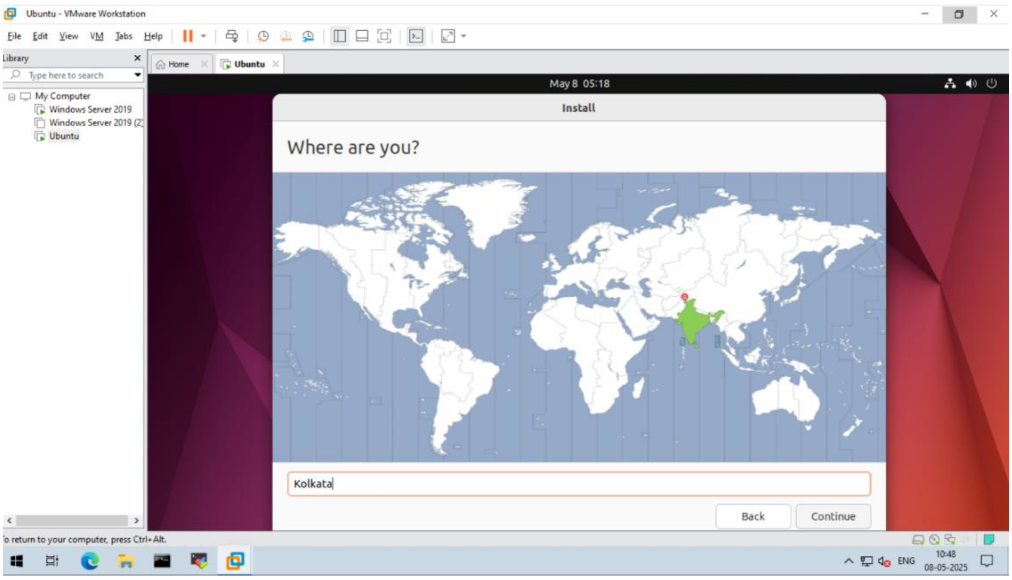


Step 12: Confirm Settings

- Click **Continue** again.

Step 13: Set Location

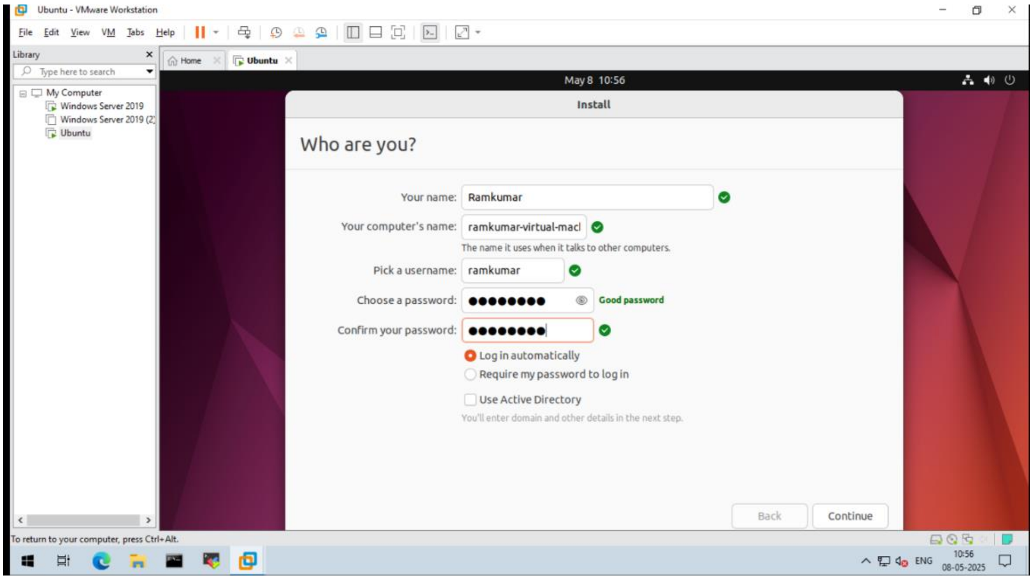
- Select your **location**.
- Click **Continue**.



Step 14: Create User Account

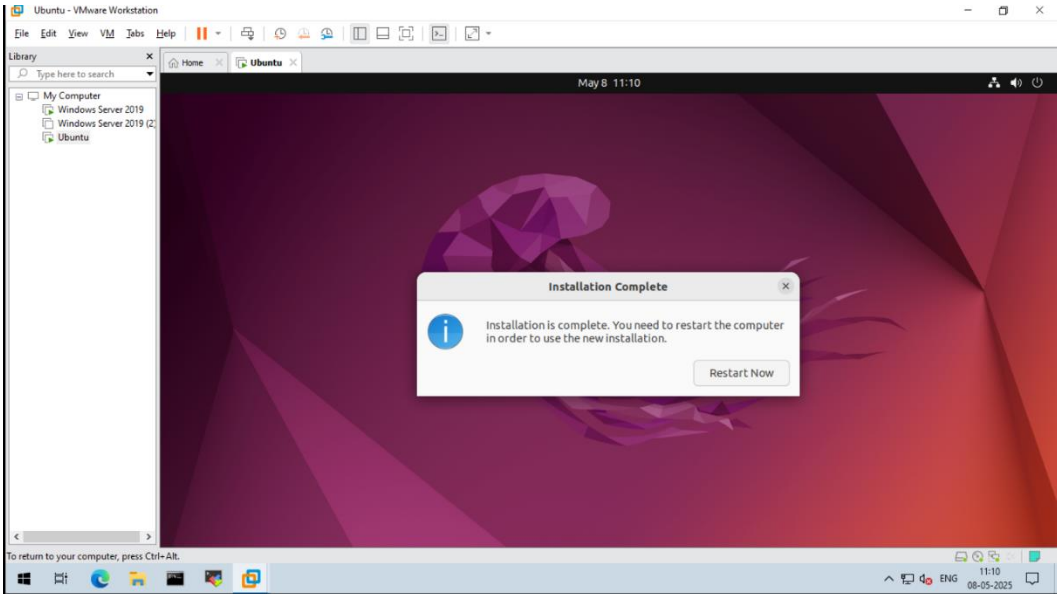
- Enter your:
  - Name
  - Computer name

    - Username
    - Password
- Select **Login Automatically**.
- Click **Continue**.



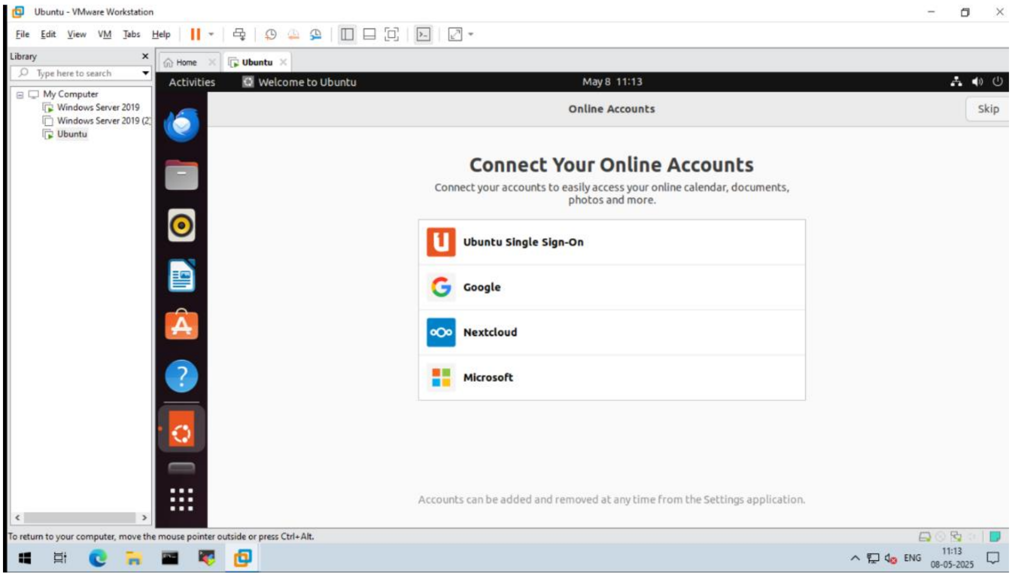
Step 15: Wait for Installation

- Wait for the installation to complete.
- Click **Restart Now** when prompted.

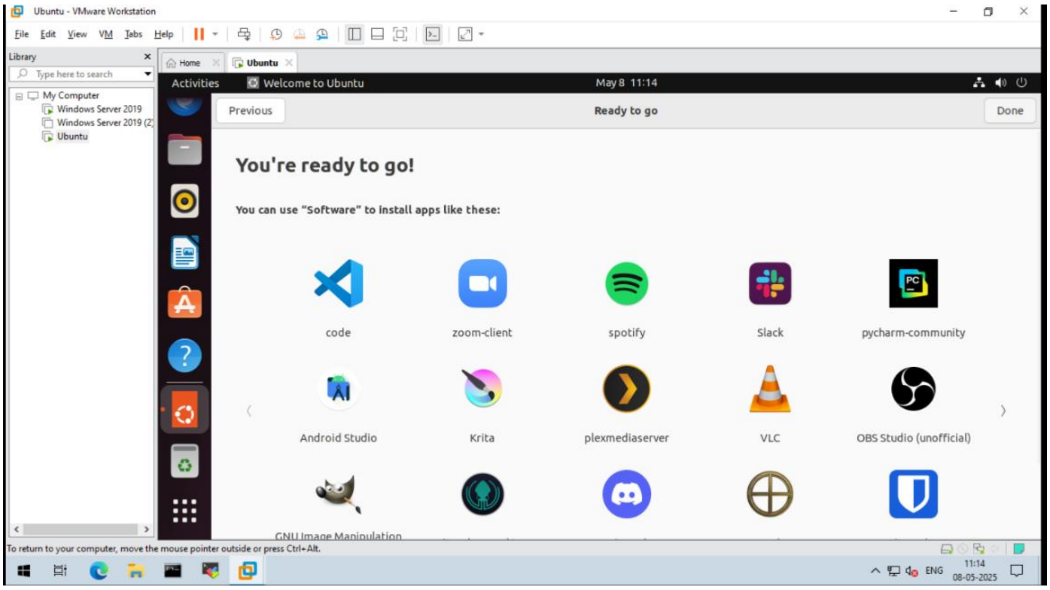


Step 16: Post-Installation Setup

- On the **Online Accounts** screen, click **Skip**.



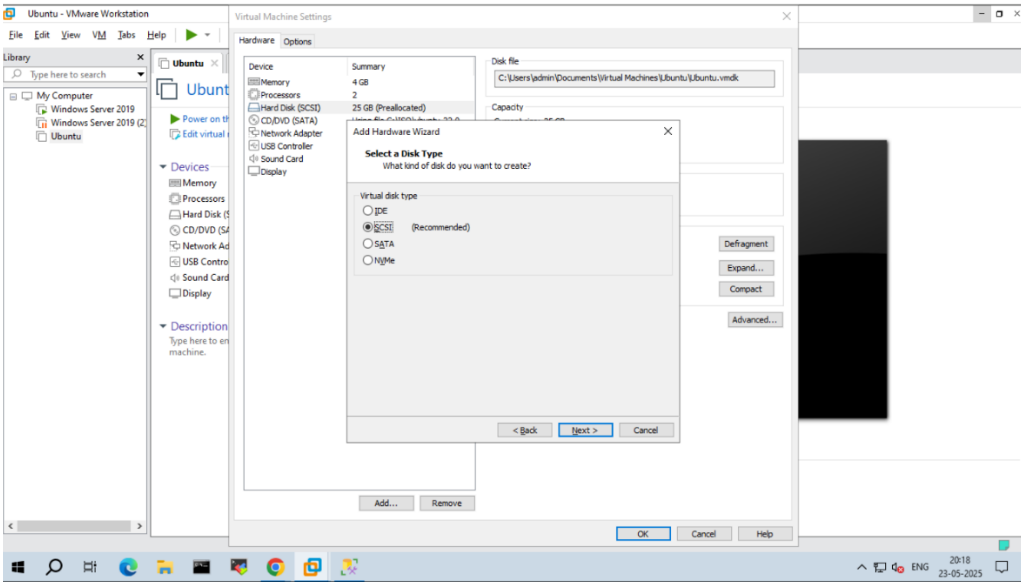
- On the next screen, click **Skip for Now**, then **Next**.
- Click **Ask Me Later**, then **Next**.
- Finally, click **Done**.



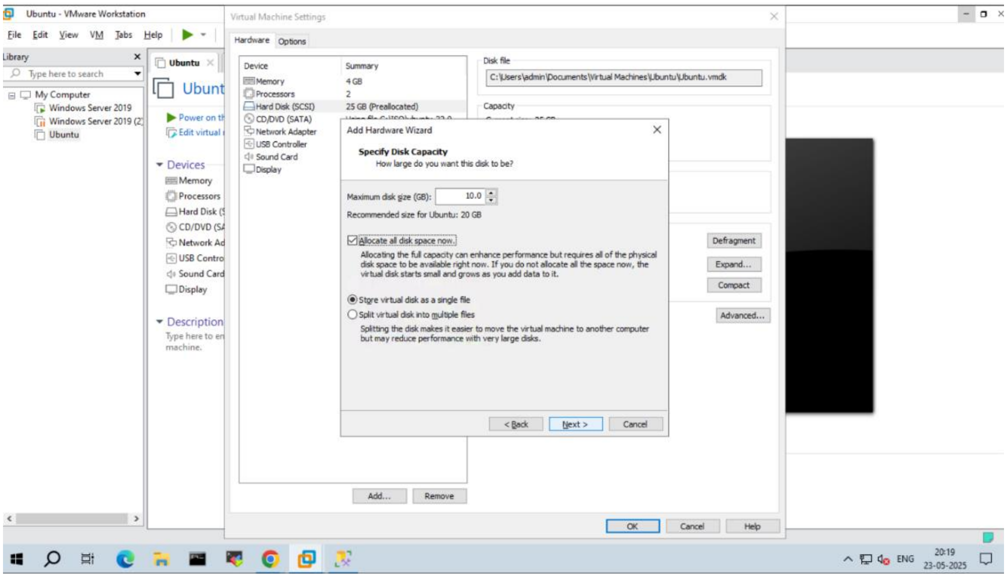
LVM Implementation – Step-by-Step Process:

Step 1: Add a 10GB Virtual Hard Disk

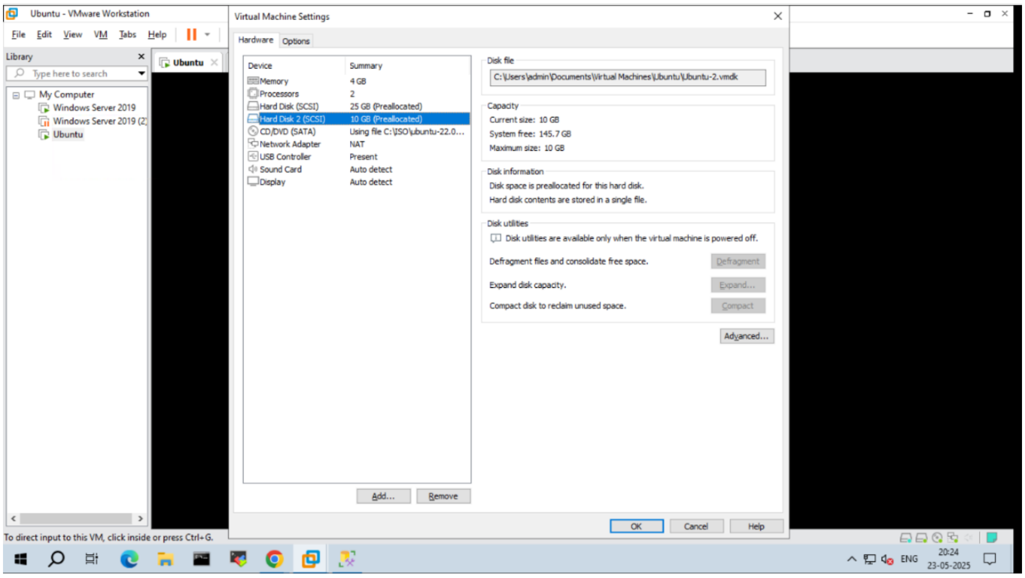
1. Shut down your virtual machine.
2. Open VM settings and click **Edit virtual machine settings**.
3. Click **Add**, choose **Hard Disk**, then click **Next**.
4. Select **SCSI** (recommended), click **Next**, and choose **Create a new virtual disk**.



5. Specify disk size as **10 GB** and click **Next**, then **Finish**.

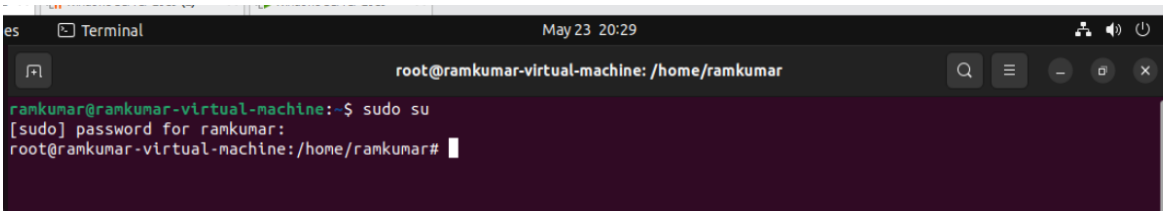


6. Click **OK** and power on the virtual machine.



Step 2:

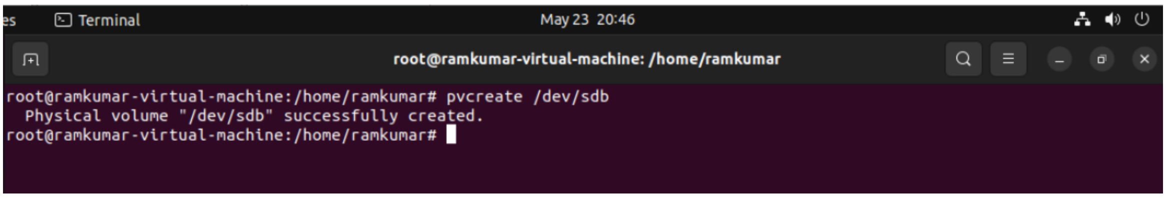
- Open the terminal. Switch to super user by using the command **"sudo su".**



1. **To Create the Physical Volume.**

   Run the command:

- **pvcreate /dev/sdb**



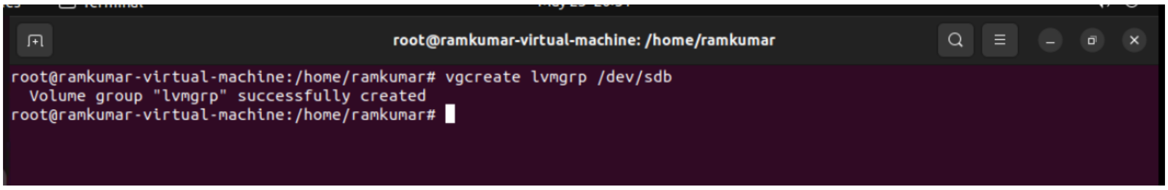
2. If prompted to install lvm2, follow the on-screen instructions and confirm with Y.
3. Verify the physical volume was created or not by running the below command:

- **pvdisplay**

Step 3: Create Volume Group

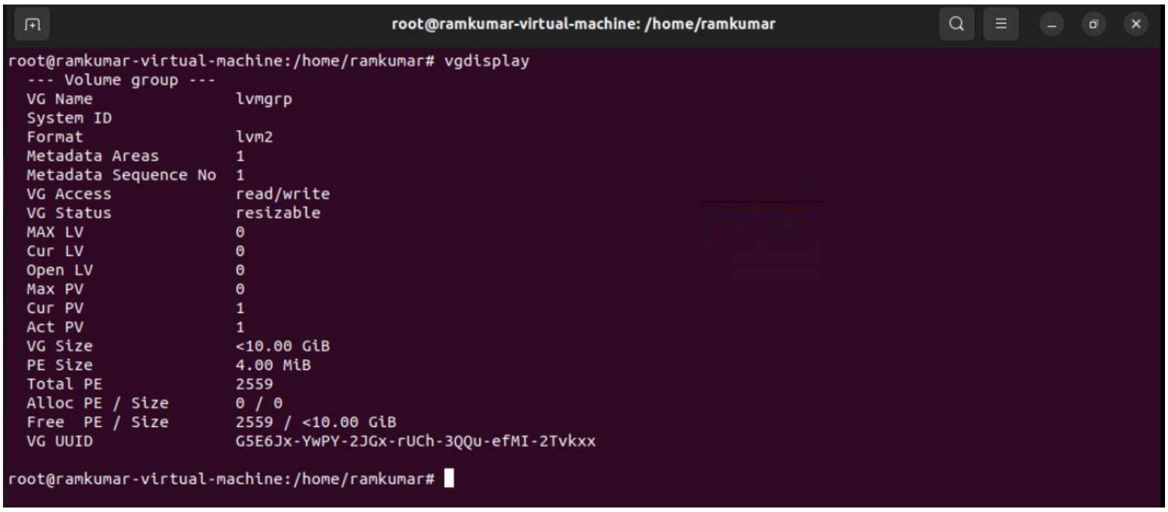
1. Create a volume group named lvmgrp by using the below command:

➢ **vgcreate lvmgrp /dev/sdb**



2. To verify whether the volume group is created or not use the below command:

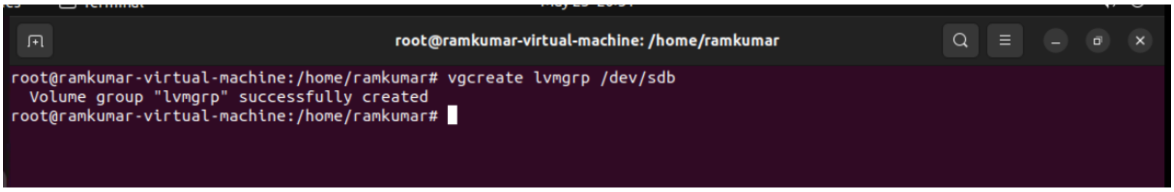
➢ **Vgdisplay**



Step 4: Create First Logical Volume (lv1 - 3GB)

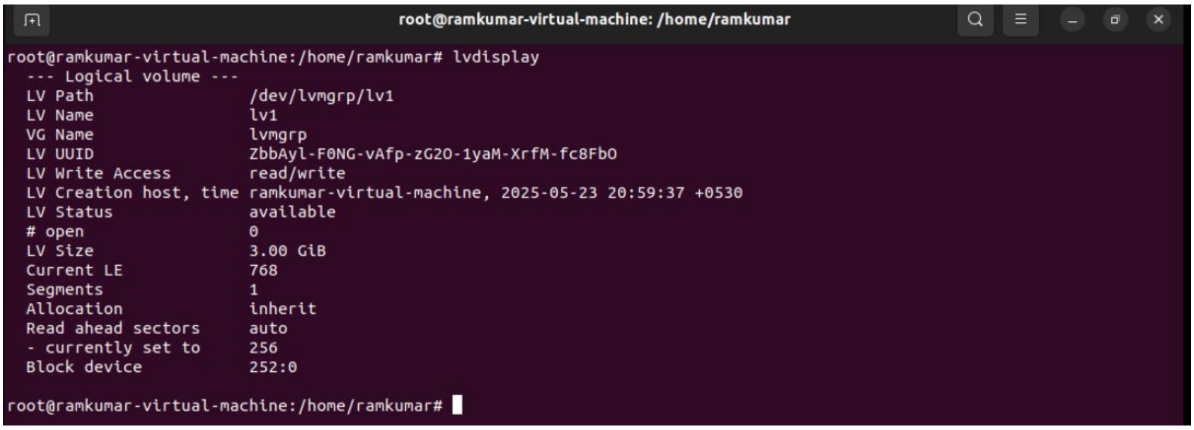
1. Then,Create the first logical volume of 3GB(lv1) by using the below command:

➢ **lvcreate -n lv1 -L 3G lvmgrp**



2. To Check whether its created or not use the below command:

➢ **vdisplay**



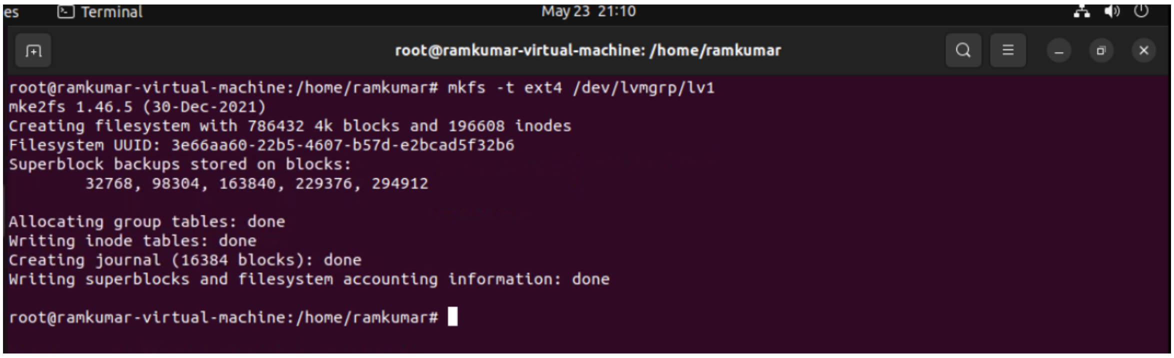
Step 5: Format and Mount lv1

1. Create a directory using the below command:

➢ **mkdir /dirlv1**

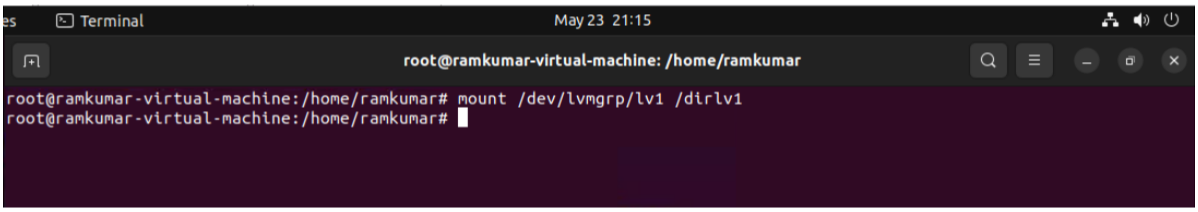
2. Format the volume by using the below command:

➢ **mkfs.ext4 /dev/lvmgrp/lv1**



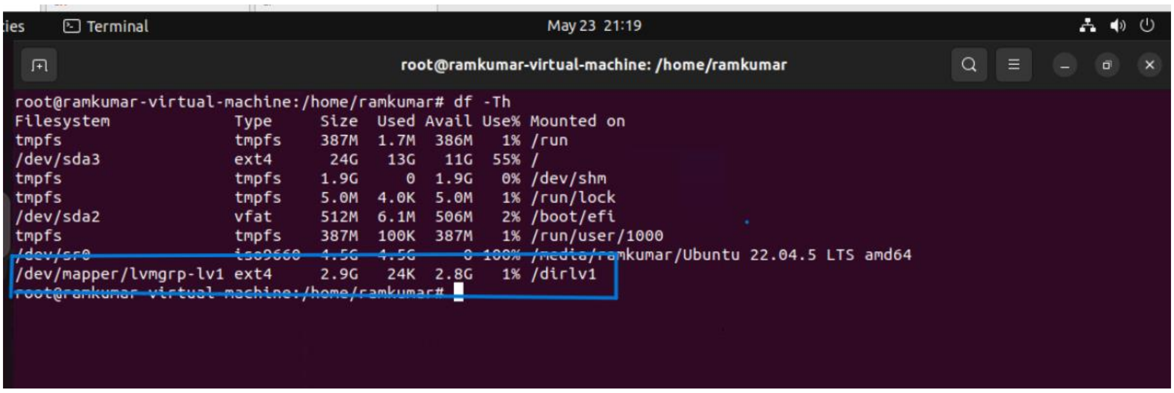
3. Then,mount it by using the below command:

➢ **mount /dev/lvmgrp/lv1 /dirlv1**



4. Check mounted volumes by using the below command:

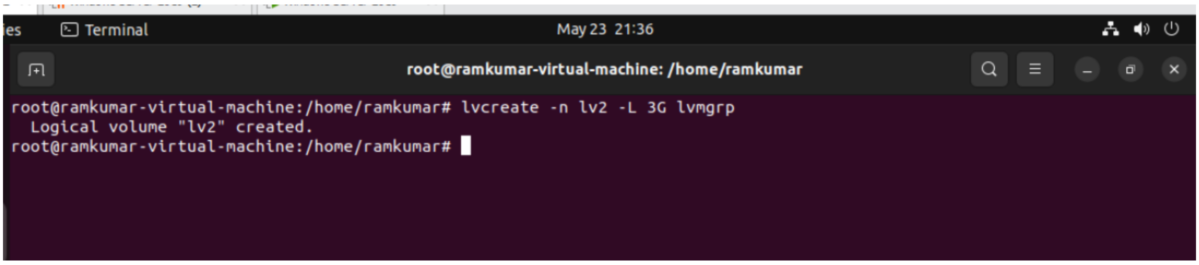
➢ df -Th



Step 6: Create Second Logical Volume (lv2 - 3GB)

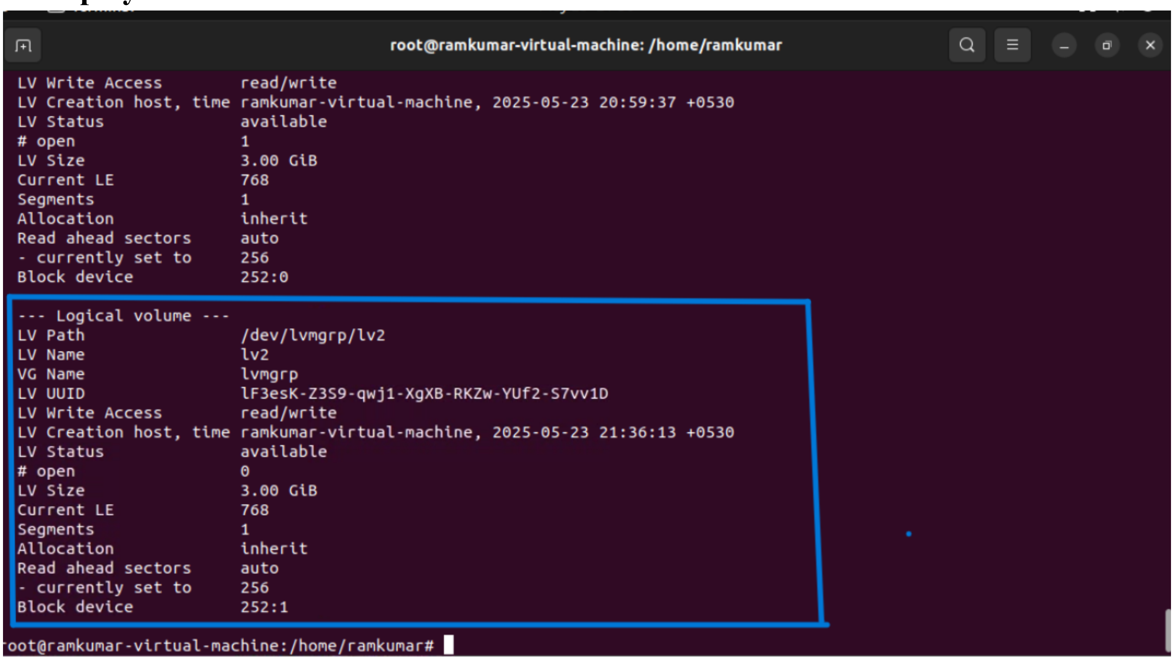
1. Create the second logical volume

➢ **lvcreate -n lv2 -L 3G lvmgrp**



2. Check if it's created

➢ **Lvdisplay**



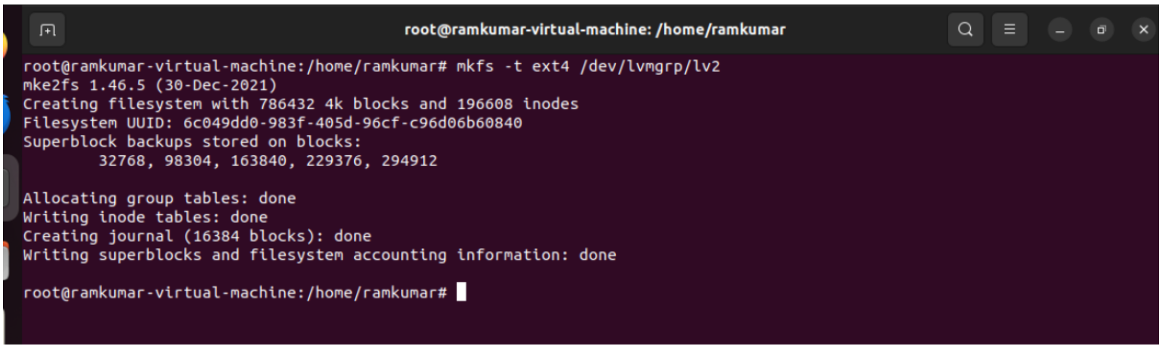
Step 7: Format and Mount lv2

1. Create mount directory

➢ **mkdir /dirlv2**

2. Format lv2

➢ **mkfs.ext4 /dev/lvmgrp/lv2**



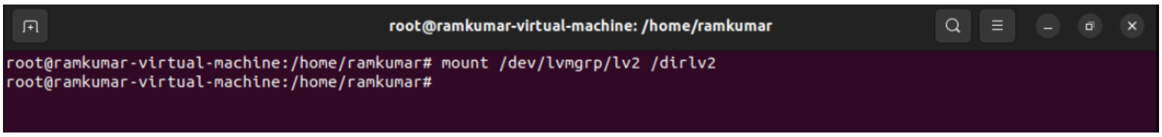
```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# mkfs -t ext4 /dev/lvmgrp/lv2
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 786432 4k blocks and 196608 inodes
Filesystem UUID: 6c049dd0-983f-405d-96cf-c96d06b60840
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ramkumar-virtual-machine:/home/ramkumar#
```

3. Mount lv2

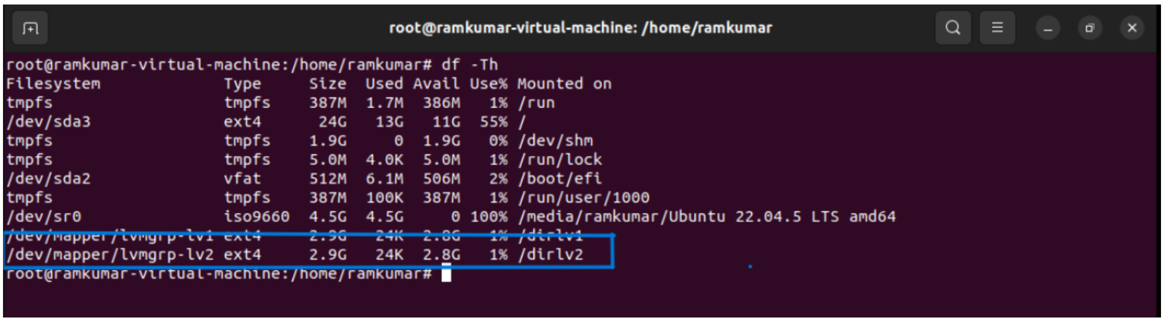
➢ **mount /dev/lvmgrp/lv2 /dirlv2**



```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# mount /dev/lvmgrp/lv2 /dirlv2
root@ramkumar-virtual-machine:/home/ramkumar#
```

4. Check mounted volumes

➢ **df -Th**



```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# df -Th
Filesystem             Type     Size  Used Avail Use% Mounted on
tmpfs                  tmpfs    387M  1.7M  386M   1% /run
/dev/sda3              ext4      24G   13G   11G  55% /
tmpfs                  tmpfs    1.9G     0  1.9G   0% /dev/shm
tmpfs                  tmpfs    5.0M  4.0K  5.0M   1% /run/lock
/dev/sda2              vfat     512M  6.1M  506M   2% /boot/efi
tmpfs                  tmpfs    387M  100K  387M   1% /run/user/1000
/dev/sr0               iso9660  4.5G  4.5G     0 100% /media/ramkumar/Ubuntu 22.04.5 LTS amd64
/dev/mapper/lvmgrp-lv1 ext4     2.9G   24K  2.8G   1% /dirlv1
/dev/mapper/lvmgrp-lv2 ext4     2.9G   24K  2.8G   1% /dirlv2
root@ramkumar-virtual-machine:/home/ramkumar#
```

Step 8: Create Third Logical Volume (lv3 - 4GB)

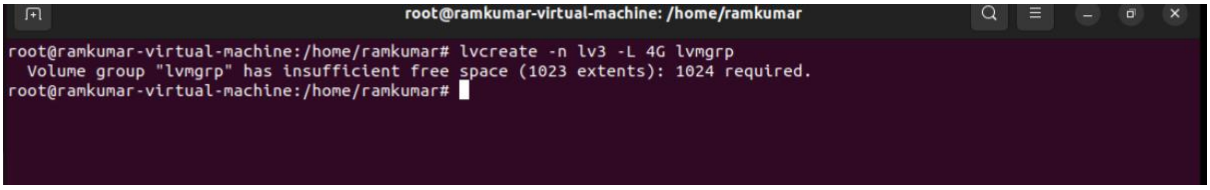
Issue While Creating Logical Volume (lv3)

When trying to create the third logical volume (lv3) using the command:

➢ **lvcreate -n lv3 -L 4G lvmgrp**

The following error was shown:

Volume group "lvmgrp" has insufficient free space (1023 extents): 1024 required.



```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# lvcreate -n lv3 -L 4G lvmgrp
  Volume group "lvmgrp" has insufficient free space (1023 extents): 1024 required.
root@ramkumar-virtual-machine:/home/ramkumar#
```

Explanation:

In LVM (Logical Volume Manager), space is divided into extents. By default, one extent = 4 MB.
To create a 4GB volume, LVM needs:

4 GB = 4096 MB

Required extents = 4096 / 4 = 1024 extents

But the volume group lvmgrp has only 1023 extents available, which is 1 extent short. That’s why LVM shows an error.

Solution:

We adjusted the logical volume size to fit within the available extents:

- **lvcreate -n lv3 -L 3.9G lvmgrp**

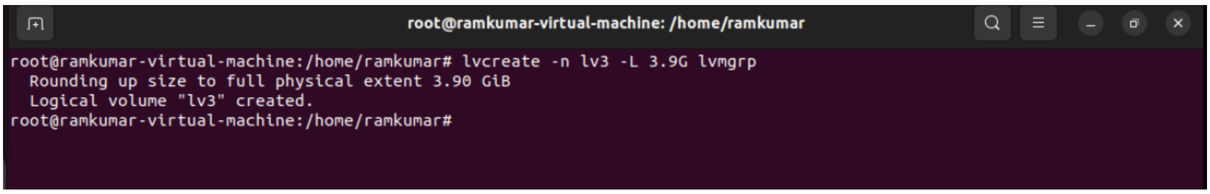
Why 3.9G?

- 3.9 GB = 3993.6 MB
- Required extents = 3993.6 / 4 ≈ 998 extents (which is less than 1023)

By creating a volume slightly smaller (3.9G), it fits into the remaining space in the volume group and avoids the error.

1. Create the third logical volume

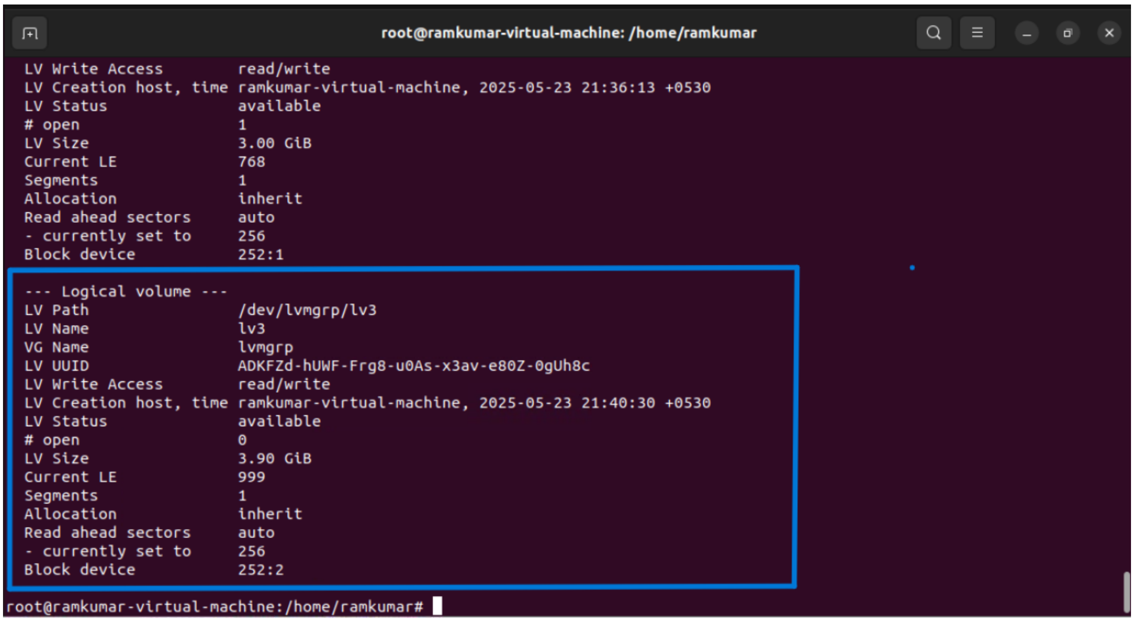
- **lvcreate -n lv3 -L 3.9G lvmgrp**



```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# lvcreate -n lv3 -L 3.9G lvmgrp
  Rounding up size to full physical extent 3.90 GiB
  Logical volume "lv3" created.
root@ramkumar-virtual-machine:/home/ramkumar#
```

2. Check if it’s created

- **Lvdisplay**



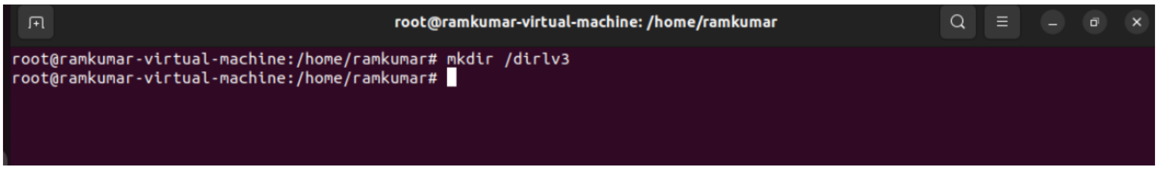
```
root@ramkumar-virtual-machine: /home/ramkumar
  LV Write Access        read/write
  LV Creation host, time ramkumar-virtual-machine, 2025-05-23 21:36:13 +0530
  LV Status              available
  # open                 1
  LV Size                3.00 GiB
  Current LE             768
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           252:1

  --- Logical volume ---
  LV Path                /dev/lvmgrp/lv3
  LV Name                lv3
  VG Name                lvmgrp
  LV UUID                ADKFZd-hUWF-Frg8-u0As-x3av-e80Z-0gUh8c
  LV Write Access        read/write
  LV Creation host, time ramkumar-virtual-machine, 2025-05-23 21:40:30 +0530
  LV Status              available
  # open                 0
  LV Size                3.90 GiB
  Current LE             999
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           252:2
root@ramkumar-virtual-machine:/home/ramkumar#
```

Step 9: Format and Mount lv3

1. Create mount directory

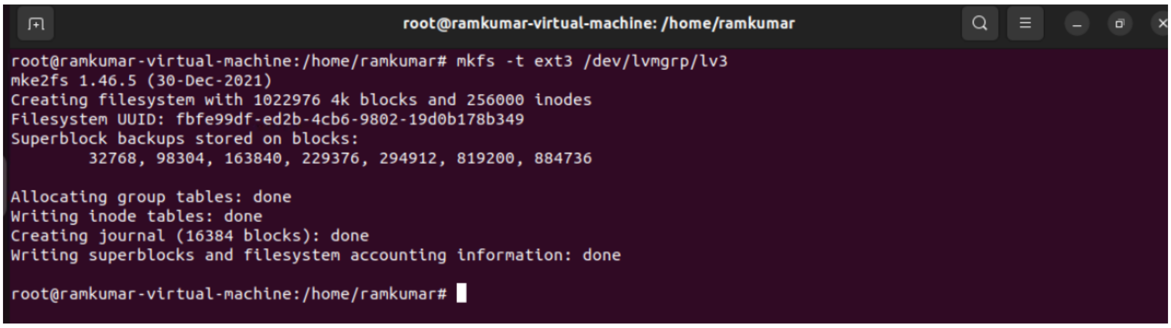
➢ **mkdir /dirlv3**



```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# mkdir /dirlv3
root@ramkumar-virtual-machine:/home/ramkumar#
```

2. Format lv3

➢ **mkfs.ext3 /dev/lvmgrp/lv3**



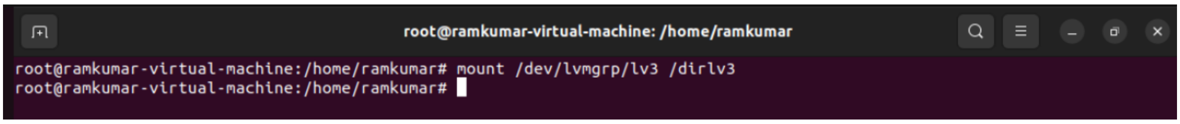
```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# mkfs -t ext3 /dev/lvmgrp/lv3
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1022976 4k blocks and 256000 inodes
Filesystem UUID: fbfe99df-ed2b-4cb6-9802-19d0b178b349
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ramkumar-virtual-machine:/home/ramkumar#
```

3. Mount lv3 using the command:

➢ **mount /dev/lvmgrp/lv3 /dirlv3**



```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# mount /dev/lvmgrp/lv3 /dirlv3
root@ramkumar-virtual-machine:/home/ramkumar#
```

4. Check mounted volumes

➢ **df -Th**

```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# df -Th
Filesystem             Type     Size  Used Avail Use% Mounted on
tmpfs                  tmpfs    387M  1.7M  386M   1% /run
/dev/sda3              ext4      24G   13G   11G  55% /
tmpfs                  tmpfs    1.9G     0  1.9G   0% /dev/shm
tmpfs                  tmpfs    5.0M  4.0K  5.0M   1% /run/lock
/dev/sda2              vfat     512M  6.1M  506M   2% /boot/efi
tmpfs                  tmpfs    387M  100K  387M   1% /run/user/1000
/dev/sr0               iso9660  4.5G  4.5G     0 100% /media/ramkumar/Ubuntu 22.04.5 LTS amd64
/dev/mapper/lvmgrp-lv1 ext4     2.9G   24K  2.8G   1% /dirlv1
/dev/mapper/lvmgrp-lv2 ext4     2.9G   24K  2.8G   1% /dirlv2
/dev/mapper/lvmgrp-lv3 ext3     3.8G   92K  3.6G   1% /dirlv3
root@ramkumar-virtual-machine:/home/ramkumar#
```

Step 10: Verify Logical Volumes and Mounts with lsblk

After creating and mounting all logical volumes, use the following command to check the block devices, volume group structure, and mount points:

➢ **lsblk**

This command shows a tree-like structure of all available block devices, their partitions, logical volumes, and where they are mounted.

This helps verify:

- The logical volumes lv1, lv2, and lv3 are created.
- They are part of lvmgrp.
- They are mounted to /dirlv1, /dirlv2, and /dirlv3.

```
root@ramkumar-virtual-machine: /home/ramkumar
root@ramkumar-virtual-machine:/home/ramkumar# lsblk
NAME         MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0          7:0    0     4K  1 loop /snap/bare/5
loop1          7:1    0  74.3M  1 loop /snap/core22/1612
loop2          7:2    0  73.9M  1 loop /snap/core22/1963
loop3          7:3    0 241.9M  1 loop /snap/firefox/6198
loop4          7:4    0 241.9M  1 loop /snap/firefox/6159
loop5          7:5    0 505.1M  1 loop /snap/gnome-42-2204/176
loop6          7:6    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop7          7:7    0  12.2M  1 loop /snap/snap-store/1216
loop8          7:8    0   568K  1 loop /snap/snapd-desktop-integration/253
loop9          7:9    0   516M  1 loop /snap/gnome-42-2204/202
loop10         7:10   0  38.8M  1 loop /snap/snapd/21759
loop11         7:11   0  12.9M  1 loop /snap/snap-store/1113
loop12         7:12   0   500K  1 loop /snap/snapd-desktop-integration/178
loop13         7:13   0  50.9M  1 loop /snap/snapd/24505
sda            8:0    0    25G  0 disk
├─sda1         8:1    0     1M  0 part
├─sda2         8:2    0   513M  0 part /boot/efi
└─sda3         8:3    0  24.5G  0 part /
sdb            8:16   0    10G  0 disk
├─lvmgrp-lv1 252:0    0     3G  0 lvm  /dirlv1
├─lvmgrp-lv2 252:1    0     3G  0 lvm  /dirlv2
└─lvmgrp-lv3 252:2    0   3.9G  0 lvm  /dirlv3
sr0           11:0    1   4.4G  0 rom  /media/ramkumar/Ubuntu 22.04.5 LTS amd64
root@ramkumar-virtual-machine:/home/ramkumar#
```

Conclusion

In this documentation, we successfully demonstrated the deployment of Ubuntu Server in a virtualized environment using VMware and implemented Logical Volume Management (LVM) for flexible disk management. Ubuntu Server provides a reliable and efficient platform for hosting services, while LVM offers powerful tools to dynamically manage storage.

Through the step-by-step process, we learned how to:

- Install Ubuntu Server in a virtual machine.
- Add a new virtual hard disk.
- Create physical and logical volumes using LVM.
- Format and mount the logical volumes for practical use.